Lecture 5

Representations and modules

Objectives. *Block I studied groups acting on sets. From now on a group acts on a vector space by linear maps, a* representation*, and this lecture asks when such an action splits into irreducible pieces. Averaging over the group answers it: if the characteristic of the field does not divide $|G|$, as over $\mathbb{C}$, every representation is a direct sum of irreducible ones (Maschke's theorem). Over $\mathbb{C}$, Schur's lemma shows that the number of times each irreducible piece occurs does not depend on the splitting, and that the irreducible representations of an abelian group are one-dimensional. Lecture 6 turns the averaging into the theory of characters.*

By the end of this lecture you will be able to:

- Decide whether a small representation is irreducible, and split $\mathbb{C}^3$ for S_3 into irreducible subrepresentations.
- Prove Maschke's theorem by averaging a projection, compute $\dim V^G$ as the average of $\operatorname{tr}\rho(g)$, and give a representation over $\mathbb{F}_2$ that is not a direct sum of irreducibles.
- Prove both forms of Schur's lemma, and deduce that an irreducible complex representation U occurs $\dim \operatorname{Hom}_G(U,V)$ times in every decomposition of V into irreducibles.
- Prove that the irreducible complex representations of a finite abelian group are one-dimensional, and list those of C_n.

5.1 Representations and modules

Throughout, G is a finite group and $\mathbb{F}$ a field; vector spaces are finite-dimensional.

Definition 5.1 (Representation)**.** A *representation* of G over $\mathbb{F}$ is a vector space V over $\mathbb{F}$ with a homomorphism $\rho\colon G \to \mathrm{GL}(V)$. We write $g \cdot v = \rho(g)v$, and say that V is a representation when ρ is understood. In a basis of V, $\rho(g)$ has a matrix $D(g)$, and $D(gh) = D(g)D(h)$.

So a representation is a linear action: an action of G on the set V (Definition 1.6) in which each $v \mapsto g \cdot v$ is linear. It is also called an $\mathbb{F}[G]$-*module*; the group algebra behind the word comes in Lecture 7.

Example 5.2 (First representations)**.**

- The *trivial representation* $\mathbf{1}$ is $\mathbb{F}$ with $g \cdot v = v$.
- The sign $\operatorname{sgn}\colon S_n \to \{\pm 1\}$ (Theorem 2.10) is a representation on $\mathbb{F}$.
- If G acts on a finite set X, it acts on $\mathbb{F}^X$, with basis $\{e_x : x \in X\}$, by $g \cdot e_x = e_{g\cdot x}$: the *permutation representation*. It is a representation, since

$$g \cdot (h \cdot e_x) = e_{g\cdot(h\cdot x)} = e_{(gh)\cdot x}.$$

The running example is S_3 on $\mathbb{C}^3$ by permuting coordinates. Its matrices, defined in § 2.5, have traces counting fixed points.

Definition 5.3 (Intertwiners). Let V and W be representations of G. An *intertwiner* is a linear map $T : V \to W$ with

$$T\,\rho_V(g) = \rho_W(g)\,T \qquad \text{for all } g \in G.$$

They form a vector space $\mathrm{Hom}_G(V, W)$, and $\mathrm{End}_G(V) = \mathrm{Hom}_G(V, V)$. An invertible intertwiner is an *isomorphism*, and then $V \cong W$.

The inverse of an invertible intertwiner is again one. In matrices, $V \cong W$ says that $D_W(g) = AD_V(g)A^{-1}$ for one invertible matrix A and every g: isomorphic representations differ by a change of basis.

Definition 5.4 (Subrepresentations). A subspace $W \subseteq V$ is a *subrepresentation* if $g \cdot W \subseteq W$ for every g. Then:

- V is *irreducible* if $V \neq 0$ and its only subrepresentations are 0 and V;
- $V = W \oplus W'$ is a *direct sum* of subrepresentations if it is a direct sum of the subspaces W, W', both subrepresentations;
- V is *indecomposable* if $V \neq 0$ and $V = W \oplus W'$ forces $W = 0$ or $W' = 0$.

The *invariants* are $V^G = \{v \in V : g \cdot v = v \text{ for all } g\}$.

An irreducible representation is indecomposable, and a one-dimensional one is irreducible. In a basis of V that begins with a basis of W, each $D(g)$ is block upper triangular; if $V = W \oplus W'$, a basis of W followed by one of W' makes it block diagonal.

Example 5.5 (The line and the plane). In $\mathbb{C}^3$, the line $L = \mathbb{C}(1, 1, 1)$ and the plane

$$E = \{x \in \mathbb{C}^3 : x_1 + x_2 + x_3 = 0\}$$

are subrepresentations: permuting the coordinates fixes $(1, 1, 1)$ and keeps their sum. So $\mathbb{C}^3$ is not irreducible. § 5.5 shows that $\mathbb{C}^3 = L \oplus E$, with E irreducible.

5.2 Averaging and Maschke

The *characteristic* of a field is the least positive integer p with $p \cdot 1 = 0$, or 0 if none exists. Averaging divides by $|G|$, so we assume the characteristic does not divide $|G|$.

Lemma 5.6 (Idempotents). *Let $P : V \to V$ be linear with $P^2 = P$. Then P is the identity on $\operatorname{im} P$, and $V = \ker P \oplus \operatorname{im} P$. If $\mathbb{F} = \mathbb{C}$, then $\operatorname{tr} P = \operatorname{rank} P$.*

Proof. For $v = Pw$ in $\operatorname{im} P$,

$$Pv = P^2 w = Pw = v.$$

Every $v \in V$ is $(v - Pv) + Pv$, with $Pv \in \operatorname{im} P$ and

$$P(v - Pv) = Pv - P^2 v = 0.$$

If $v \in \ker P \cap \operatorname{im} P$, then $v = Pv$ and $Pv = 0$. So $V = \ker P \oplus \operatorname{im} P$.

In a basis of $\operatorname{im} P$ followed by one of $\ker P$, the matrix of P is diagonal, with $\operatorname{rank} P$ entries 1 and the rest 0. So $\operatorname{tr} P = \operatorname{rank} P$. ∎

Proposition 5.7 (Averaging). *Let V be a representation of G, and*

$$P = \frac{1}{|G|} \sum_{g \in G} \rho(g).$$

(i) $\rho(h)P = P = P\rho(h)$ *for every* $h \in G$.

(ii) $P^2 = P$, *and* $\operatorname{im} P = V^G$.

(iii) If $\mathbb{F} = \mathbb{C}$, *then*

$$\dim V^G = \frac{1}{|G|} \sum_{g \in G} \operatorname{tr} \rho(g).$$

Proof. (i) As g runs over G, so does hg. Hence

$$\rho(h)P = \frac{1}{|G|} \sum_{g \in G} \rho(hg) = P,$$

and in the same way $P\rho(h) = P$.

(ii) Invariance gives $h \cdot Pv = Pv$, so $\operatorname{im} P \subseteq V^G$. If $v \in V^G$, each term of Pv is v, so $Pv = v$. So $\operatorname{im} P = V^G$, and $P^2 = P$.

(iii) By Lemma 5.6,

$$\dim V^G = \operatorname{rank} P = \operatorname{tr} P = \frac{1}{|G|} \sum_{g \in G} \operatorname{tr} \rho(g). \qquad \blacksquare$$

Example 5.8 (Permutation representations). On $\mathbb{C}^X$, $\operatorname{tr} \rho(g) = |\mathrm{Fix}(g)|$, and a vector $\sum_x a_x e_x$ is invariant exactly when a is constant on each orbit. So $\dim(\mathbb{C}^X)^G$ is the number of orbits. The averaging formula is Burnside's lemma (Theorem 1.20) again. For S_3 on $\mathbb{C}^3$,

$$\dim(\mathbb{C}^3)^{S_3} = \frac{1}{6}(3 + 1 + 1 + 1 + 0 + 0) = 1,$$

the invariants are L, and every entry of P is $\frac{1}{3}$:

$$Px = \frac{1}{3}(x_1 + x_2 + x_3)\,(1, 1, 1).$$

Averaging also produces intertwiners. For representations U and V, G acts on the linear maps $\mathrm{Hom}(U, V)$ by

$$g \cdot T = \rho_V(g)\, T\, \rho_U(g)^{-1}.$$

This is a representation, since

$$\begin{aligned} g \cdot (h \cdot T) &= \rho_V(g)\rho_V(h)\, T\, \rho_U(h)^{-1}\rho_U(g)^{-1} \\ &= (gh) \cdot T, \end{aligned}$$

and $g \cdot T = T$ says $\rho_V(g)T = T\rho_U(g)$. So

$$\mathrm{Hom}(U, V)^G = \mathrm{Hom}_G(U, V).$$

Theorem 5.9 (Maschke). *Let V be a representation of G over $\mathbb{F}$, where the characteristic of $\mathbb{F}$ does not divide $|G|$.*

(i) Every subrepresentation W of V has a complement that is a subrepresentation: $V = W \oplus W'$.

(ii) V *is a direct sum of irreducible subrepresentations (the empty sum if* $V = 0$*).*

(iii) If $\mathbb{F} = \mathbb{C}$*,* V *has an inner product with* $\langle g \cdot u, g \cdot v \rangle = \langle u, v \rangle$ *for all* g, u, v*. For it, each* $\rho(g)$ *is unitary, and the orthogonal complement of a subrepresentation is a subrepresentation.*

Proof. (i) Choose a linear projection π of V onto W: $\operatorname{im} \pi = W$, and π is the identity on W. Average it in $\operatorname{Hom}(V, V)$:

$$\pi_0 = \frac{1}{|G|} \sum_{g \in G} \rho(g)\, \pi\, \rho(g)^{-1}.$$

An intertwiner. By Proposition 5.7(ii), $\pi_0 \in \operatorname{Hom}(V, V)^G = \operatorname{End}_G(V)$.

A projection onto W. Each term maps V into $g \cdot W = W$, and fixes each $w \in W$, as π fixes $\rho(g)^{-1} w \in W$. So π_0 is an idempotent with image W, and $V = W \oplus \ker \pi_0$ by Lemma 5.6.

The complement. $\ker \pi_0$ is a subrepresentation: if $\pi_0 v = 0$, then

$$\pi_0(g \cdot v) = g \cdot \pi_0 v = 0.$$

(ii) By induction on $\dim V$. If V is 0 or irreducible, there is nothing to prove. Otherwise $V = W \oplus W'$ by (i), with $W \neq 0, V$, and the decompositions of W and W' together decompose V.

(iii) Average any inner product $(\cdot, \cdot)$ on V:

$$\langle u, v \rangle = \frac{1}{|G|} \sum_{g \in G} (g \cdot u, g \cdot v).$$

It is sesquilinear, and $\langle v, v \rangle > 0$ for $v \neq 0$, as every term is.

It is invariant, as gh runs over G with g; so each $\rho(g)$ is unitary.

If W is a subrepresentation, $v \in W^\perp$ and $w \in W$, then $g^{-1} \cdot w \in W$, and

$$\langle w, g \cdot v \rangle = \left\langle g^{-1} \cdot w, v \right\rangle = 0.$$

So $g \cdot v \in W^\perp$.[1] ∎

Example 5.10 (Guard: the characteristic divides $|G|$)**.** Let $C_2 = \{e, g\}$ act on $\mathbb{F}_2^2$ in the standard basis by

$$D(g) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Since $1 + 1 = 0$, $D(g)^2 = \mathbb{1}$. The three lines are spanned by e_1, e_2, $e_1 + e_2$. The first is invariant; g exchanges the other two. Thus the representation is indecomposable but not irreducible: Maschke's splitting fails. Here $|G| = 0$ in $\mathbb{F}_2$. Lecture 11 needs every representation to split, and works over $\mathbb{C}$.

5.3 Schur's lemma and multiplicities

Henceforth $\mathbb{F} = \mathbb{C}$, except in one guard. A field is *algebraically closed* if every non-constant polynomial over it has a root in it.

Theorem 5.11 (Schur's lemma)**.** *Let* U *and* U' *be irreducible representations of* G*.*

(i) Over any field, an intertwiner $T : U \to U'$ *is* 0 *or an isomorphism.*

(ii) Over an algebraically closed field $\mathbb{F}$*, such as* $\mathbb{C}$*,* $\operatorname{End}_G(U) = \mathbb{F}\mathbb{1}$*. Hence* $\dim \operatorname{Hom}_G(U, U')$ *is* 1 *if* $U \cong U'$*, and* 0 *otherwise.*

[1] Serre's scalar product is antilinear in the second slot.

Proof. (i) The kernel and image of T are subrepresentations: if $Tu = 0$, then

$$T(g \cdot u) = g \cdot Tu = 0,$$

and $g \cdot Tu = T(g \cdot u)$ lies in $\operatorname{im} T$.

If $T \neq 0$, then $\ker T \neq U$, so $\ker T = 0$; and $\operatorname{im} T \neq 0$, so $\operatorname{im} T = U'$. So T is bijective.

(ii) Let $T \in \operatorname{End}_G(U)$. As $U \neq 0$ and $\mathbb{F}$ is algebraically closed, T has an eigenvalue c. Then $T - c\mathbb{1}$ is an intertwiner that is not injective. The kernel-and-image result forces $T = c\mathbb{1}$.

If $S\colon U \to U'$ is an isomorphism, then $T \mapsto S^{-1}T$ is a linear bijection from $\operatorname{Hom}_G(U, U')$ onto $\operatorname{End}_G(U) = \mathbb{F}\,\mathbb{1}$. If $U \not\cong U'$, the same result leaves only $T = 0$. ∎

Example 5.12 (Guard: Schur's lemma over $\mathbb{R}$)**.** Let $C_4 = \langle g \rangle$ act on $\mathbb{R}^2$ by the rotation through $90°$,

$$g \mapsto J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Its characteristic polynomial $t^2 + 1$ has no real root, so no line is invariant, and $\mathbb{R}^2$ is irreducible over $\mathbb{R}$.

But J commutes with every $\rho(g^k) = J^k$, so $J \in \operatorname{End}_{C_4}(\mathbb{R}^2)$, and J is not a scalar: (ii) fails over $\mathbb{R}$. Solving $XJ = JX$ gives

$$\operatorname{End}_{C_4}(\mathbb{R}^2) = \{a\mathbb{1} + bJ : a, b \in \mathbb{R}\}, \qquad J^2 = -\mathbb{1}$$

a copy of $\mathbb{C}$.

Over $\mathbb{C}$, J has two eigenlines,

$$J(1, -\mathrm{i}) = \mathrm{i}\,(1, -\mathrm{i}), \qquad J(1, \mathrm{i}) = -\mathrm{i}\,(1, \mathrm{i})$$

and $\mathbb{C}^2$ splits. Lecture 9 returns to real representations.

By Maschke's theorem, $V = U_1 \oplus \cdots \oplus U_k$ with irreducible U_j. Does the number of U_j isomorphic to U depend on the decomposition?

Corollary 5.13 (Multiplicities)**.** *Let $V = U_1 \oplus \cdots \oplus U_k$ with each U_j irreducible, and let U be irreducible. Then*

$$\#\{j : U_j \cong U\} = \dim \operatorname{Hom}_G(U, V).$$

So this number, the multiplicity $m_U(V)$ *of U in V, does not depend on the decomposition. Moreover $V \cong W$ if and only if $m_U(V) = m_U(W)$ for every irreducible U.*

Proof. Let $\iota_j\colon U_j \to V$ be the inclusion and $p_j\colon V \to U_j$ the projection along the other summands. Both intertwine, since each summand is invariant. The maps

$$T \mapsto (p_1 T, \ldots, p_k T),$$
$$(T_1, \ldots, T_k) \mapsto \sum_j \iota_j T_j$$

are inverse linear bijections between $\operatorname{Hom}_G(U, V)$ and $\operatorname{Hom}_G(U, U_1) \oplus \cdots \oplus \operatorname{Hom}_G(U, U_k)$. By scalar Schur,

$$\dim \operatorname{Hom}_G(U, V) = \sum_j \dim \operatorname{Hom}_G(U, U_j) = \#\{j : U_j \cong U\}.$$

If $S\colon V \to W$ is an isomorphism, $T \mapsto ST$ is a linear bijection $\operatorname{Hom}_G(U, V) \to \operatorname{Hom}_G(U, W)$, so the multiplicities agree. Conversely, if they agree, decompose V and W, pair their summands isomorphically, and take the direct sum of the pairings. ∎

5.4 Abelian groups

Corollary 5.14 (Abelian groups)**.**

(i) Every irreducible complex representation of a finite abelian group is one-dimensional.

(ii) For every complex representation ρ of a finite group G and every $g \in G$, $\rho(g)$ is diagonalisable, and its eigenvalues are $o(g)$-th roots of unity.

Proof. (i) Let U be irreducible, and $h \in G$. As $hg = gh$ for all g, $\rho(h)$ commutes with every $\rho(g)$: it lies in $\mathrm{End}_G(U)$, so it is a scalar by Theorem 5.11(ii). Then every line in U is a subrepresentation, and as U is irreducible, $\dim U = 1$.

(ii) The same ρ is a representation of the abelian group $\langle g \rangle$. By Maschke's theorem and (i), V is a direct sum of lines invariant under g, and in a basis of these lines $\rho(g)$ is diagonal. Each eigenvalue c satisfies $c^{o(g)} = 1$, as

$$\rho(g)^{o(g)} = \rho(e) = \mathbb{1}.$$

■

Over $\mathbb{R}$, (i) fails: Example 5.12 is a two-dimensional irreducible representation of C_4.

Example 5.15 (The cyclic group)**.** Let $C_n = \langle g \rangle$ and $\zeta = \exp(2\pi \mathrm{i}/n)$. For $j = 0, 1, \ldots, n-1$,

$$\rho_j(g^k) = \zeta^{jk}$$

is a one-dimensional representation: it is well defined, as $\zeta^{jn} = 1$, and

$$\rho_j(g^k g^l) = \zeta^{j(k+l)} = \rho_j(g^k)\rho_j(g^l).$$

Every one-dimensional ρ has $\rho(g)^n = 1$, so $\rho(g) = \zeta^j$ for one j, and $\rho = \rho_j$. Two one-dimensional representations are isomorphic only if equal, since $a\,\rho(g)\,a^{-1} = \rho(g)$. By one-dimensionality, C_n has exactly n irreducibles up to isomorphism.

5.5 Worked example

Split $\mathbb{C}^3$ for S_3 into irreducibles, and meet an irreducible plane for D_4.

The splitting. Each $x \in \mathbb{C}^3$ splits as

$$x = \frac{1}{3}(x_1 + x_2 + x_3)(1,1,1) + \big(x - \frac{1}{3}(x_1 + x_2 + x_3)(1,1,1)\big),$$

the first term in L, the second in E, as its coordinates sum to 0. And $L \cap E = 0$, as $1+1+1 \neq 0$. So $\mathbb{C}^3 = L \oplus E$.

The matrices on E. Take the basis $f_1 = e_1 - e_2$, $f_2 = e_2 - e_3$ of E. Then

$$(1\,2) \cdot f_1 = e_2 - e_1 = -f_1,$$

and, applying the same transposition to f_2,

$$(1\,2) \cdot f_2 = e_1 - e_3 = f_1 + f_2.$$

For the three-cycle,

$$(1\,2\,3) \cdot f_1 = e_2 - e_3 = f_2,$$

while

$$(1\,2\,3) \cdot f_2 = e_3 - e_1 = -f_1 - f_2.$$

and the columns of the matrices are these images:

$$D_E((1\,2)) = \begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad D_E((1\,2\,3)) = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}.$$

E is irreducible. A subrepresentation other than 0 and E would be a line $\mathbb{C}v$ invariant under $(1\,2)$ and $(1\,2\,3)$: a common eigenvector. $D_E((1\,2))$ has the two eigenvalues -1 and 1, so its only eigenlines are $\mathbb{C}f_1$ and $\mathbb{C}(f_1 + 2f_2)$, where $f_1 + 2f_2 = (1, 1, -2)$. But

$$(1\,2\,3) \cdot f_1 = f_2, \qquad (1\,2\,3) \cdot (f_1 + 2f_2) = -2f_1 - f_2$$

and neither lies on the line it came from. So E is irreducible, and $\mathbb{C}^3 = \mathbf{1} \oplus E$: by Corollary 5.13, $\mathbf{1}$ and E have multiplicity 1 in $\mathbb{C}^3$, and sgn has multiplicity 0.

D_4 on $\mathbb{C}^2$. Let

$$R = \begin{pmatrix} \mathrm{i} & 0 \\ 0 & -\mathrm{i} \end{pmatrix}, \qquad S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Then $R^4 = S^2 = \mathbb{1}$ and $SRS^{-1} = R^{-1}$. Each element of D_4 is $r^k s^j$ for unique $k \in \{0, 1, 2, 3\}$ and $j \in \{0, 1\}$ (Example 1.13), and the relations $r^4 = s^2 = e$ and $sr = r^{-1}s$ give the product rule

$$r^a s^b \cdot r^c s^d = r^{a+(-1)^b c} s^{b+d},$$

with exponents of r mod 4, of s mod 2. R and S satisfy the same relations, so the same computation gives

$$R^a S^b \cdot R^c S^d = R^{a+(-1)^b c} S^{b+d},$$

and $\rho(r^k s^j) = R^k S^j$ is a representation of D_4.

It is irreducible: R has the distinct eigenvalues i and $-\mathrm{i}$, so its only eigenlines are $\mathbb{C}e_1$ and $\mathbb{C}e_2$, and S swaps them.

Schur's lemma at work: r^2 is central in D_4, as

$$sr^2s^{-1} = r^{-2} = r^2,$$

so $\rho(r^2) \in \mathrm{End}_{D_4}(\mathbb{C}^2)$, and indeed

$$\rho(r^2) = R^2 = -\mathbb{1}.$$

Lecture 8 builds this representation from $C_4 = \langle r \rangle$, and Lecture 9 computes its Frobenius–Schur indicator.

Self-check. Compute $\dim E^{S_3}$ from Proposition 5.7(iii). *Answer:* substituting the traces of the matrices above,

$$\dim E^{S_3} = \frac{1}{6}\big(2 + 3 \cdot 0 + 2 \cdot (-1)\big) = 0,$$

as it must be for an irreducible $E \ncong \mathbf{1}$.

Summary.

- *The main results:* if the characteristic does not divide $|G|$, every representation is a direct sum of irreducibles (Theorem 5.9); over $\mathbb{C}$, by Schur's lemma (Theorem 5.11), U occurs $\dim \mathrm{Hom}_G(U, V)$ times (Corollary 5.13), and abelian groups have one-dimensional irreducibles (Corollary 5.14).
- *Handed on:* $\dim V^G$ by averaging (Proposition 5.7) and $\mathrm{Hom}(U, V)^G = \mathrm{Hom}_G(U, V)$, with which Lecture 6 computes $\dim \mathrm{Hom}_G(U, V)$; the multiplicity space $\mathrm{Hom}_G(U, V)$ to Lecture 7; D_4 on $\mathbb{C}^2$ to Lectures 8 and 9.
- *Next:* Lecture 6 asks how much of a representation its trace remembers.

References

Serre, *Linear Representations of Finite Groups*: definitions, §§1.1–1.3, pp. 3–5; the averaged projection, Thm 1, and the invariant scalar product, p. 6; the direct sum of irreducibles, Thm 2, p. 7; Schur's lemma, Prop. 4 and its corollaries, pp. 13–14. James and Liebeck, *Representations and Characters of Groups*: Maschke's theorem over $\mathbb{R}$ and $\mathbb{C}$, Thm 8.1, p. 70, and its failure over $\mathbb{F}_p$, Example 8.2(2), p. 71;[2] Schur's lemma, 9.1, p. 78; abelian groups, Prop. 9.5, p. 81. Etingof et al., *Introduction to Representation Theory*: Maschke's theorem over a field whose characteristic does not divide $|G|$, Thm 4.1.1, p. 59, and its converse, Prop. 4.1.2, p. 60.

[2] James and Liebeck use right modules, $v \mapsto vg$, so their matrices are the transposes of ours.